

Kadane's Algorithm

Difficulty: Medium | Accuracy: 36.28% | Submissions: 1.3M | Points: 4 | Average Time: 20m

You are given an integer array `arr[]`. You need to find the maximum sum of a subarray (containing at least one element) in the array `arr[]`.

Examples

Input: `arr[] = [2, 3, -8, 7, -1, 2, 3]`

Output: 11

Explanation: The subarray `[7, -1, 2, 3]` has the largest sum 11.

Input: `arr[] = [-2, -4]`

Output: -2

Explanation: The subarray `[-2]` has the largest sum -2.

Input: `arr[] = [5, 4, 1, 7, 8]`

Output: 25

Explanation: The subarray `[5, 4, 1, 7, 8]` has the largest sum 25.

Constraints

$1 \leq \text{arr.size()} \leq 10^5$

$-10^4 \leq \text{arr}[i] \leq 10^4$

Topic Tags

Arrays, Dynamic Programming

Company Tags

Zoho, Flipkart, Morgan Stanley, Accolite, Amazon, Microsoft, Samsung, Snapdeal, 24*7 Innovation Labs, Citrix, D-E-Shaw, FactSet, Hike, Housing.com, MetLife, Ola Cabs, Oracle, Payu, Teradata, Visa, Walmart, Adobe, Google, Arcesium